Battery literature digest — week of 02 November 2026

9 papers selected from 410 scanned. Scores out of 10; click a title to open the paper.

1. Discovering Underpotentially Deposited Crystalline Sodium Metal Nanoparticles on the Surface of Hard Carbon Anodes

8/10 · *Journal of the American Chemical Society* · cryo-EM and cryo-electron tomography

Cryo-ET combined with cryo-EM localizes crystalline Na metal clusters in sodiated hard carbon in 3D. They sit on or near particle surfaces, not in closed bulk nanopores as commonly assumed. This changes the picture of underpotential Na plating and points to surface and interface mitigation.

2. Diagnosing the Role of Mn Additives in Aqueous Zn/MnO2 Batteries

7/10 · *Advanced Energy Materials* · operando X-ray fluorescence (Mn2+ tracking)

Operando XRF directly tracks electrolyte Mn2+ concentration during cycling of Zn/MnO2 cells. It suggests the Mn additive works mainly as extra redox-active Mn rather than by suppressing dissolution. It also points to incomplete MnO2 dissolution and Mn2+ redeposition as a cause of fade, which the additive does not fix.

3. Operando Mapping of Population Phase Separation and Lithium-Plating Dynamics in Graphite Electrode Using Optical Microscopy and Microstructure Modeling

7/10 · *Advanced Energy Materials* · operando optical microscopy + 3D phase-field modeling

Combines operando optical microscopy of graphite staging across the particle population with microstructure-resolved phase-field simulation. It proposes a plating mechanism driven by interfacial ohmic resistance that sets in before surface saturation. It also uses particle-size distribution and temperature as probes of phase-separation pathways and plating onset.

4. Operando X-ray Photoelectron Spectroscopy Reveals Oxide Solid Electrolyte Stability against Electrochemically Plated Lithium

7/10 · *ACS Energy Letters* · lab-based operando XPS during electrochemical Li plating

Electrochemical Li plating inside the XPS avoids the artefacts of evaporated or sputtered Li. LATP is reduced to a Li+/Ti3+ mixed-conducting interphase that partly relaxes, while LLZO shows no resolvable Zr or La reduction. This is a clean, direct test of the thermodynamic-instability predictions. It is spectroscopy, not spatially resolved imaging.

5. Mechanistic Insights of Jahn–Teller Effect Suppression in High-Entropy Prussian Blue Analogue Cathodes

6/10 · *Journal of the American Chemical Society* · EXAFS + machine-learning interatomic potential

EXAFS shows no Mn–N bond-length splitting in a high-entropy PBA, only a broader distribution. MLIP calculations explain this as randomly oriented [MnN6] distortions along x, y and z that average out and break cooperative J-T order. The paper links this mechanism to 91% capacity retention over 50,000 cycles. It is local-structure work, not imaging.

6. Vacancy-Induced Ligand-Hole Character Redirects Oxygen Redox in Li-Rich Layered Oxides

6/10 · *Journal of the American Chemical Society* · mapping RIXS (mRIXS), soft X-ray absorption, DFT

Oxygen-vacancy preconditioning in quasi-single-crystal Li-rich oxide is linked, via mapping RIXS, to earlier oxygen participation and more delocalised O holes in hybridised Ni–O states. This suppresses Mn reduction and voltage decay. The spectroscopy is solid, but it is bulk-averaged and not spatially resolved, and the work is still a material-modification study.

7. Mechanistic Insights into Sodium Storage in N-Doped Graphene/MoS2 Heterostructures Revealed by Operando Raman Spectroscopy

6/10 · *ACS Nano* · operando spectroelectrochemical Raman

Uses three-atomic-layer G/NG/MoS2 stacks so operando Raman can separate charge-doping from strain during Na+ cycling. Concludes storage is mainly reversible accumulation in interlayer and blister-like reservoirs formed by solvent cointercalation. The outer layer controls ion access: N-doped graphene gives up to 40× the capacity of pristine graphene, and MoS2 on top blocks infiltration.

8. Watching Electrochemical Interfaces in Motion with High-Speed Atomic Force Microscopy

6/10 · *ACS Nano* · operando high-speed AFM (electrochemical HS-AFM)

Perspective on applying sub-second HS-AFM to electrochemical interfaces (electrodeposition, dissolution, corrosion, lithiation morphology, nanobubbles). Relevant as an emerging operando nanoscale imaging method, but it is a perspective, broader than batteries, and lithiation is only one topic.

9. Operando Raman Spectroscopy-Guided 24-Hour Aging Assessment of Activated Carbon Positive Electrodes for Lithium-Ion Capacitors

6/10 · *ACS Energy Letters* · operando Raman spectroscopy (D1 band analysis)

Uses operando Raman during a 24-h high-voltage cycling protocol on activated carbon cathodes. The turning point in D1 band width (narrowing from CEI/gas passivation, then broadening from corrosion) correlates with capacity decay across different ACs. This gives a fast aging diagnostic, though it is bulk Raman rather than spatially resolved.